ENGINEERING SOCIETY · TECHNICAL TEAM

Technical Blueprint

The full data model, the real technology stack, and why each decision was made. This is what we build against, locked.

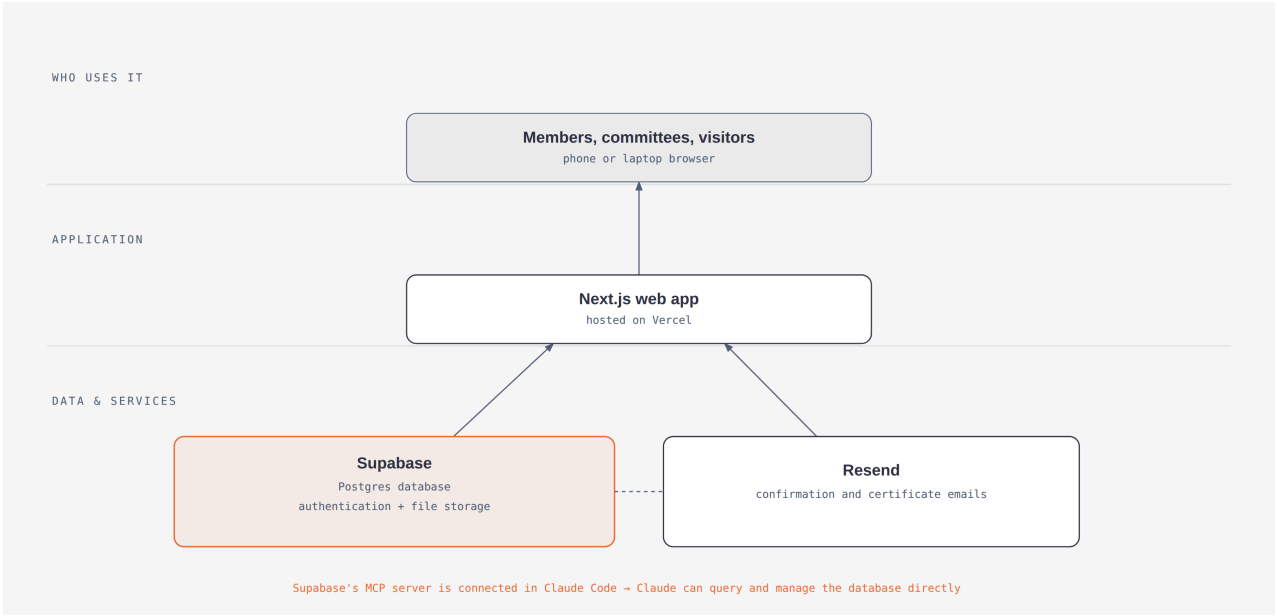
Version 1.0 · Prepared by Hisham Alhinai, Technical Team Lead

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PART 1 · THE STACK

The real technology, top to bottom

Two pieces, chosen to keep a small, mostly non-technical team productive with AI assistance, without needing to build infrastructure from scratch.



Architecture decisions

Short and written down, so the reasoning survives even if the person who made the call isn't in the room. This stack replaces an earlier version of this document, which chose Cloudflare Pages, D1, R2, and Better Auth for cost and independence from any single vendor. That reasoning was sound and is kept below for the record, it wasn't a mistake, it was superseded by a better fit for how this team actually works.

ADR-05: Next.js on Vercel

Decision: Build the member-facing site and forms in Next.js, deployed on Vercel. Supersedes ADR-01 (Cloudflare Pages).

Why: Vercel is built by the same team as Next.js, so deployment, previews, and framework updates stay friction-free. Its free tier comfortably covers a society-sized platform.

ADR-06: Supabase for database, auth, and storage

Decision: Use Supabase (hosted Postgres, authentication, and file storage) instead of Cloudflare D1, R2, and Better Auth. Supersedes ADR-02 and ADR-03.

Why: Supabase has a native MCP server, which means Claude, in this chat and in Claude Code, can inspect the schema, run queries, and manage auth and storage directly. For a team that's mostly non-technical and leaning on AI assistance, that direct connection is a bigger practical advantage than the cost savings and vendor independence Cloudflare offered. Supabase's free tier still keeps us at effectively no cost at our current scale.

ADR-04: Resend for email

Decision: Send confirmation and certificate emails through Resend rather than a mail server we manage. Unchanged from the earlier version of this document.

Why: registration and certificate emails (UC-01, UC-04) need to arrive reliably. Resend's free tier covers our current volume, and running our own mail server is real infrastructure work with no learning value for the team.

Considered and set aside: our own VPS

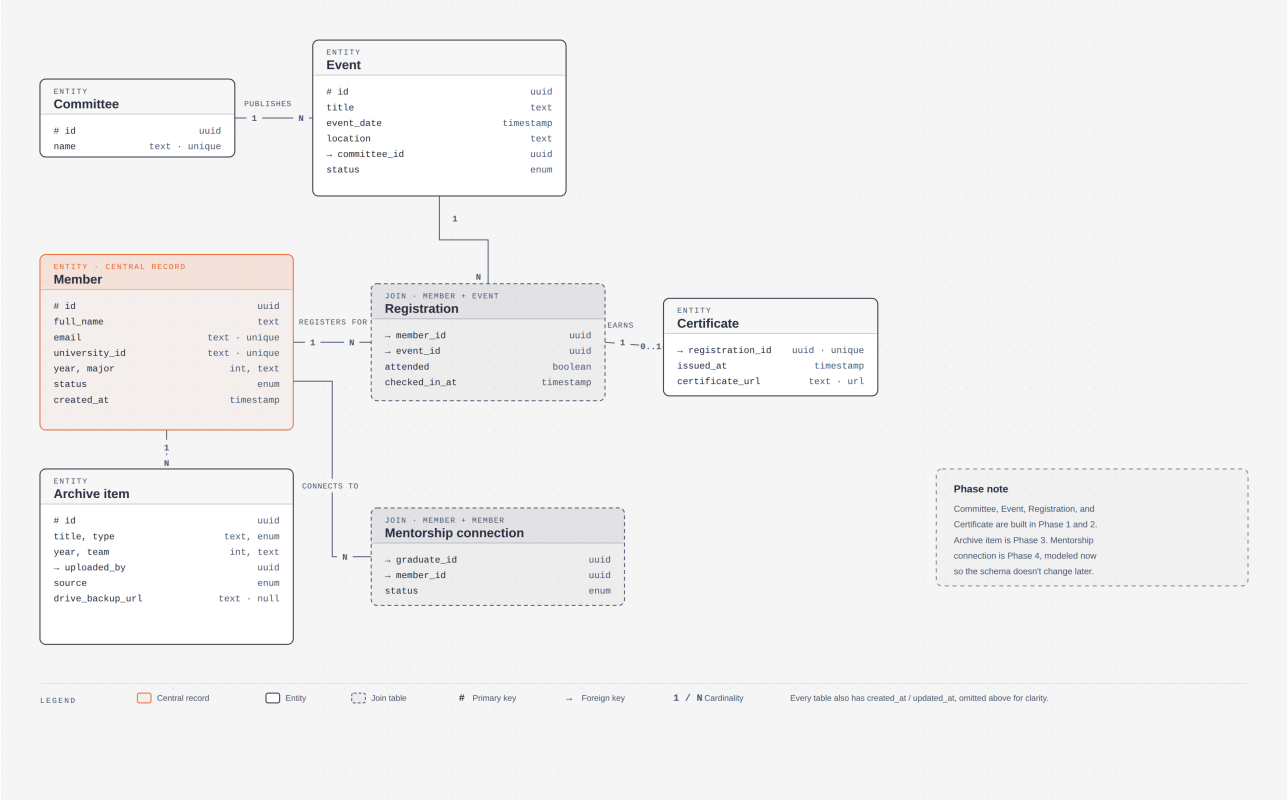
We looked at hosting everything ourselves on our VPS instead.

Why not, for now: a VPS gives full control, but it also makes us responsible for security patches, backups, and uptime, real sysadmin work with no safety net if something breaks at 2am before an event. Both Cloudflare and Supabase remove that responsibility for effectively no cost. If we outgrow the free tier or need something neither can do, the VPS option stays open, this isn't a permanent door closing.

PART 2 · THE DATA MODEL

Full data model, v1.0

Every entity, every field, every connection, covering all seven approved use cases. This is the technical version of the simplified diagram we walked through as a team, same shape, full detail.



Seven entities. Committee, Event, Registration, and Certificate are built in Phase 1 and 2. Archive item is Phase 3. Mentorship connection is Phase 4, modeled now so the schema doesn't need to change later, but not built until then.

Field notes worth knowing

Every unique field is enforced by the database.

email and university_id on Member, name on Committee. This is what makes UC-01's duplicate detection reliable, the database itself refuses a second row with the same email, we don't have to catch it in application code.

Registration is the busiest table in the system.

Every event registration, every attendance check-in, and indirectly every certificate all pass through this one table. If something's wrong with attendance or certificates, this is the first place to look.

Certificate points to Registration, not to Member or Event directly.

That one link already knows which member and which event, since Registration links both. Storing member_id or event_id again on Certificate would just be the same fact written twice, and the two copies could quietly disagree with each other over time.

Archive item's drive_backup_url is nullable on purpose.

Items uploaded straight to the platform (UC-05) never had a Drive link. Only items migrated from Drive (UC-06) fill this field in. Nullable here means "not every row needs this," not "we forgot to fill it in."